An Analysis of BLEU score

BLEU score

BLEU score -- Part 1

When r ≤ c {\displaystyle r\leq c} , the brevity penalty B P = 1 {\displaystyle BP=1} , meaning that we don't punish long candidates, and only punish short candidates. When r > c {\displaystyle r>c} , the brevity penalty B P = e 1 − r / c {\displaystyle BP=e^{1-r/c}}

BLEU score -- Part 2

Modified n-gram precision Define the modified n-gram precision function to be p n (S ^ ; S) := ∑ i = 1 M ∑ s ∈ G n (y ^ (i)) min (C (s , y ^ (i)) , max y ∈ S i C (s , y)) ∑ i = 1 M ∑ s ∈ G n (y ^ (i)) C (s , y ^ (i)) {\displaystyle p_{n}({\hat {S}};S):={\frac {\sum _{i=1}^{M}\sum _{s\in G_{n}({\hat {y}}^{(i)})}\min(C(s,{\hat {y}}^{(i)}),\max _{y\in S_{i}}C(s,y))}{\sum _{i=1}^{M}\sum _{s\in G_{n}({\hat {y}}^{(i)})}C(s,{\hat {y}}^{(i)})}}} The modified n-gram, which looks complicated, is merely a straightforward generalization of the prototypical case: one candidate sentence and one reference sentence. In this case, it is p n ({ y ^ } ; { y }) = ∑ s ∈ G n (y ^) min (C (s , y ^) , C (s , y)) ∑ s ∈ G n (y ^) C (s , y ^) {\displaystyle p_{n}(\{{\hat {y}}\};\{y\})={\frac {\sum _{s\in G_{n}({\hat {y}})}\min(C(s,{\hat {y}}),C(s,y))}{\sum _{s\in G_{n}({\hat {y}})}C(s,{\hat {y}})}}} To work up to this expression, we start with the most obvious n-gram count summation: ∑ s ∈ G n (y ^) C (s , y) = number of n-substrings in y ^ that appear in y {\displaystyle \sum _{s\in G_{n}({\hat {y}})}C(s,y)={\text{number of n-substrings in }}{\hat {y}}{\text{ that appear in }}y}

BLEU score -- Part 3

as the word 'the' and the word 'cat' appear once each in the candidate, and the total number of words is two. The modified bigram precision would be 1 / 1 {\displaystyle 1/1} as the bigram, "the cat" appears once in the candidate. It has been pointed out that precision is usually twinned with recall to overcome this problem [7], as the unigram recall of this example would be 3 / 6 {\displaystyle 3/6} or 2 / 7 {\displaystyle 2/7} . The problem being that as there are multiple reference translations, a bad translation could easily have an inflated recall, such as a translation which consisted of all the words in each of the references.[8] To produce a score for the whole corpus, the modified precision scores for the segments are combined using the geometric mean multiplied by a brevity penalty to prevent very short candidates from receiving too high a score. Let r be the total length of the reference corpus, and c the total length of the translation corpus. If c ≤ r {\displaystyle c\leq r} , the brevity penalty applies, defined to be e (1 − r / c) {\displaystyle e^{(1-r/c)}} . (In the case of multiple reference sentences, r is taken to be the sum of the lengths of the sentences whose lengths are closest to the lengths of the candidate sentences. However, in the version of the metric used by NIST evaluations prior to 2009, the shortest reference sentence had been used instead.) iBLEU is an interactive version of BLEU that allows a user to visually examine the BLEU scores obtained by the candidate translations. It also allows comparing two different systems in a visual and interactive manner which is useful for system development.[9]

BLEU score -- Part 4

where m {\displaystyle ~m} is number of words from the candidate that are found in the reference, and w t {\displaystyle ~w_{t}} is the total number of words in the candidate. This is a perfect score, despite the fact that the candidate translation above retains little of the content of either of the references. The modification that BLEU makes is fairly straightforward. For each word in the candidate translation, the algorithm takes its maximum total count, m m a x {\displaystyle ~m_{max}} , in any of the reference translations. In the example above, the word "the" appears twice in reference 1, and once in reference 2. Thus m m a x = 2 {\displaystyle ~m_{max}=2} . For the candidate translation, the count m w {\displaystyle m_{w}} of each word is clipped to a maximum of m m a x {\displaystyle m_{max}} for that word. In this case, "the" has m w = 7 {\displaystyle ~m_{w}=7} and m m a x = 2 {\displaystyle ~m_{max}=2} , thus m w {\displaystyle ~m_{w}} is clipped to 2. These clipped counts m w {\displaystyle ~m_{w}} are then summed over all distinct words in the candidate. This sum is then divided by the total number of unigrams in the candidate translation. In the above example, the modified unigram precision score would be:

BLEU score -- Part 5

c {\displaystyle c} is the length of the candidate corpus, that is, c := ∑ i = 1 M | y ^ (i) | {\displaystyle c:=\sum _{i=1}^{M}|{\hat {y}}^{(i)}|} where | y | {\displaystyle |y|} is the length of y {\displaystyle y} .

BLEU score -- Part 6

Basic setup A basic, first attempt at defining the BLEU score would take two arguments: a candidate string y ^ {\displaystyle {\hat {y}}} and a list of reference strings (y (1) , . . . , y (N)) {\displaystyle (y^{(1)},...,y^{(N)})} . The idea is that B L E U (y ^ ; y (1) , . . . , y (N)) {\displaystyle BLEU({\hat {y}};y^{(1)},...,y^{(N)})} should be close to 1 when y ^ {\displaystyle {\hat {y}}} is similar to y (1) , . . . , y (N) {\displaystyle y^{(1)},...,y^{(N)}} , and close to 0 if not. As an analogy, the BLEU score is like a language teacher trying to score the quality of a student translation y ^ {\displaystyle {\hat {y}}} by checking how closely it follows the reference answers y (1) , . . . , y (N) {\displaystyle y^{(1)},...,y^{(N)}} . Since in natural language processing, one should evaluate a large set of candidate strings, one must generalize the BLEU score to the case where one has a list of M candidate strings (called a "corpus") (y ^ (1) , ■ , y ^ (M)) {\displaystyle ({\hat {y}}^{(1)},\cdots ,{\hat {y}}^{(M)})} , and for each candidate string y ^ (i) {\displaystyle {\hat {y}}^{(i)}} , a list of reference candidate strings S i := (y (i , 1) , . . . , y (i , N i)) {\displaystyle S_{i}:=(y^{(i,1)},...,y^{(i,N_{i})})} . Given any string y = y 1 y 2 ■ y K {\displaystyle y=y_{1}y_{2}\cdots y_{K}} , and any integer n ≥ 1 {\displaystyle n\geq 1} , we define the set of its n-grams to be G n (y) = { y 1 ■ y n , y 2 ■ y n + 1 , ■ , y K − n + 1 ■ y K } {\displaystyle G_{n}(y)=\{y_{1}\cdots y_{n},y_{2}\cdots y_{n+1},\cdots ,y_{K-n+1}\cdots y_{K}\}} Note that it is a set of unique elements, not a multiset allowing redundant elements, so that, for example, G 2 (a b a b) = { a b , b a } {\displaystyle G_{2}(abab)=\{ab,ba\}} . Given any two strings s , y {\displaystyle s,y} , define the substring count C (s , y) {\displaystyle C(s,y)} to be the number of appearances of s {\displaystyle s} as a substring of y {\displaystyle y} . For example, C (a b , a b c b a b) = 2 {\displaystyle C(ab,abcbab)=2} . Now, fix a candidate corpus S ^ := (y ^ (1) , ■ , y ^ (M)) {\displaystyle {\hat {S}}:=({\hat {y}}^{(1)},\cdots ,{\hat {y}}^{(M)})} , and reference candidate corpus S = (S 1 , ■ , S M) {\displaystyle S=(S_{1},\cdots ,S_{M})} , where each S i := (y (i , 1) , . . . , y (i , N i)) {\displaystyle S_{i}:=(y^{(i,1)},...,y^{(i,N_{i})})} .